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United States Patent Application Publication

20250276549

Kind Code

A1

Publication Date

September 04, 2025

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APPARATUSES, SYSTEMS, AND METHODS FOR DETERMINING AND VERIFYING OPERATIONAL STATES OF FIFTH WHEELS

Abstract

A method of tracking operational state of a fifth wheel includes sensing, with a sensor, movement of a movable component of the fifth wheel that moves as the fifth wheel locks onto a kingpin of a towed vehicle and determining, with a controller having a memory, an operational state of the fifth wheel based on the sensed movement of the movable component. The method also includes logging the operational state of the fifth wheel on the memory such that operation and/or wear of the fifth wheel is observable over time.

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Appl. No.: **19/211986**

Filed: **May 19, 2025**

Related U.S. Application Data

parent US continuation 18390974 20231220 parent-grant-document US 12304259 child US 19211986

parent US continuation 18052687 20221104 parent-grant-document US 11865882 child US 18390974

parent US continuation 16776058 20200129 parent-grant-document US 11524536 child US 18052687

us-provisional-application US 62805679 20190214

Publication Classification

Int. Cl.: B60D1/58 (20060101); B60D1/01 (20060101); G01D5/14 (20060101)

U.S. Cl.:

CPC B60D1/58 (20130101); B60D1/015 (20130101); G01D5/142 (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to and is a continuation of U.S. application Ser. No. 18/390,974, filed Dec. 20, 2023, which is a continuation of U.S. application Ser. No. 18/052,687, filed Nov. 4, 2022, now U.S. Pat. No. 11,865,882 issued Jan. 9, 2024, which is a continuation of U.S. application Ser. No. 16/776,058, filed Jan. 29, 2020, now U.S. Pat. No. 11,524,536 issued Dec. 13, 2022, which claims priority to U.S. Provisional Patent Application No. 62/805,679 filed Feb. 14, 2019, are incorporated by reference herein in their entireties.

FIELD

[0002] The present disclosure relates to fifth wheels, and specifically to determining and verifying operational states of the fifth wheels.

BACKGROUND

[0003] The following U.S. Patents are incorporated herein by reference in entirety.

[0004] U.S. Pat. No. 5,516,138 discloses a mechanism for locking and unlocking of a kingpin of a fifth wheel. The mechanism includes a jaw member, a wedge member, a bumper member and a lever member interconnecting the jaw member, the wedge and the bumper member. A handle member includes a handle extension member.

[0005] U.S. Pat. No. 5,641,174 discloses an interconnection of the jaw, wedge and operating handle in a fifth wheel facilitates the provision of an indicator on a secondary locking mechanism which further ensures the security of the system. The jaw is connected to the operating handle by a pivoting timing lever which pivots off a pin on the jaw such that the jaw remains engaged with the fifth wheel until the wedge is substantially removed from engagement with the jaw.

[0006] U.S. Pat. No. 7,735,849 discloses a fifth wheel hitch, a locking mechanism for retaining a trailer kingpin within a fifth wheel slot. The locking mechanism includes a jaw assembly comprised of two opposing jaw members pivotally attached at one end to the underside of the hitch plate, a longitudinally sliding cam interposed between the jaw members with a tip that contacts a bumper. The bumper is pivotally attached to tie bar that has its rear most end pivotally attached to the underside of the hitch plate. The mechanism also includes a wedge member and a secondary lock member pivotally attached thereto, where the lock member has a guide extension inserted through a guide hole in the tie bar.

[0007] U.S. Pat. No. 8,210,558 discloses a secondary lock assembly for a fifth wheel, where the fifth wheel includes a hitch plate with a rearward opening slot to receive a trailer kingpin and a transversely sliding primary locking member for retaining the kingpin within the slot. The assembly comprises a tie bar pivotally connected at its middle to the primary locking member and a transversely oriented pull bar pivotally connected at an inner end to the forward end of the tie bar. The pull bar comprises a rearward offset tab. A latch is pivotally connected roughly at the center of the latch to the forward end of the tie bar.

[0008] U.S. Pat. No. 9,302,557 discloses a fifth wheel includes a top plate having a throat that is adapted to receive a kingpin of a trailer. The fifth wheel is equipped with a locking mechanism including a jaw slidably connected to the top plate and slidable between a closed position where the jaw blocks passage of a kingpin out of the throat of the fifth wheel and an open position where a

kingpin may pass into and out of the throat of the fifth wheel. The jaw has an edge adapted to engage a kingpin positioned in the throat of the fifth wheel when the jaw is in the closed position. [0009] U.S. Pat. No. 9,327,782 discloses a fifth wheel includes a top plate having a throat that is adapted to receive a kingpin of a trailer. The fifth wheel is equipped with a locking mechanism including a jaw slidably connected to the top plate and slidable between a closed position where the jaw blocks passage of a kingpin out of the throat of the fifth wheel and an open position where a kingpin may pass into and out of the throat of the fifth wheel. The jaw has an edge adapted to engage a kingpin positioned in the throat of the fifth wheel when the jaw is in the closed position. [0010] U.S. Pat. No. 9,738,333 discloses a fifth wheel includes a top plate having a throat that is adapted to receive a kingpin of a trailer. A pair of locking jaws are pivotally connected to the top plate and pivotal between a closed configuration where the pair of locking jaws block passage of a kingpin out of the throat and the locking jaws are held primarily in compression and an open configuration where a kingpin may pass into and out of the throat.

SUMMARY

[0011] This Summary is provided to introduce a selection of concepts that are further described below in the Detailed Disclosure. This Summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

[0012] In certain examples, a method for determining an operational state of a fifth wheel includes sensing, with at least one sensor, magnetic flux caused by a magnet on a movable component movable to lock the fifth wheel to a kingpin of a towed vehicle, determining an end position of the movable component based on the magnetic flux, comparing the end position of the movable component to a threshold position, and determining an operational state of the fifth wheel based on the comparison of the end position of the movable component to the threshold position.

[0013] In certain examples, a fifth wheel system includes a fifth wheel configured to couple to a kingpin of a towed trailer and the fifth wheel has a movable component movable to lock the fifth wheel to the kingpin. A magnet is coupled to the movable component, and a sensor is in operative association with the fifth wheel and configured to sense magnetic flux caused by the magnet and generate data corresponding to the magnetic flux as the movable component moves to lock the fifth wheel to the kingpin. A controller is configured to receive the data and process the data to determine an end position of the movable component as the movable component moves, and the controller is further configured to compare the end position of the movable component to a threshold position to thereby determine operational state of the fifth wheel.

[0014] In certain examples, a method of verifying an operational state of a fifth wheel includes sensing a position of a movable component on a fifth wheel movable to lock the fifth wheel to a kingpin of a towed vehicle, determining a locked state of the fifth wheel based on the sensed position, and monitoring the position of the movable component for a predetermined amount of time after detecting the locked state. If a threshold change in the sensed position of the movable component is detected within the predetermined amount of time, the method includes storing a positive indicator of manual verification of the locked state of the fifth wheel by an operator.

[0015] In certain examples, a fifth wheel system includes a fifth wheel configured to couple to a kingpin of a towed trailer and the fifth wheel having a movable component movable to lock the fifth wheel to the kingpin. A sensor senses a position of the movable component, and a controller is configured to determine a locked state of the fifth wheel based on the position of the movable component and monitor the position of the movable component for a predetermined amount of time after detecting the locked state and detect a threshold change in the position of the movable component within the predetermined amount of time. The controller then stores a positive indicator of manual verification of the locked state of the fifth wheel by an operator.

[0016] Various other features, objects, and advantages will be made apparent from the following description taken together with the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] The present disclosure is described with reference to the following Figures. The same numbers are used throughout the Figures to reference like features and like components.

[0018] FIG. 1 is a bottom view of an example fifth wheel. An operating arm is shown in a locked position such that a kingpin is locked in the fifth wheel. A pull handle is shown retracted into the fifth wheel.

[0019] FIG. 2 is a view like FIG. 1 with the operating arm in an unlocked position such that the kingpin can be removed or inserted into the fifth wheel. The pull handle is shown in an extended position and extending from the fifth wheel.

[0020] FIG. 3 is an enlarged view within line 3-3 on FIG. 2. A pawl member is near a stop surface and is adjacent to a linear series of three sensors.

[0021] FIG. 4 is a view like FIG. 3 with the pawl member in close proximity to a first sensor and the indicator emitting light having a first color.

[0022] FIG. 5 is a view like FIG. 3 with the pawl member in close proximity to a second sensor and the indicator emitting light having a second color.

[0023] FIG. 6 depicts an example method of the present disclosure for verifying an operational state of the fifth wheel.

[0024] FIG. 7 is a schematic of an example sensing system of the present disclosure.

[0025] FIG. 8A-8B are example graphical representation of potential positions along which a movable component of the fifth wheel may move during the operation of the fifth wheel.

[0026] FIG. 9 depicts an example method of the present disclosure for determining an operational state of the fifth wheel.

DETAILED DISCLOSURE

[0027] It is known to connect a towed trailer to a towing vehicle via a connection assembly commonly referred to as a fifth wheel. Specifically, a fifth wheel is a primary locking assembly on the towing vehicle that engages a kingpin of the towed trailer to thereby securely couple the towing vehicle to the towed trailer. Fifth wheels are constructed to avoid/prevent inadvertent disengagement of the kingpin from the fifth wheel.

[0028] FIGS. 1-2 are bottom or underside views of an example fifth wheel 10 of the present disclosure. The fifth wheel 10 has a top plate 12, a flange 13, and a throat 14 into which a kingpin 16 of a towed trailer (not shown) is received. The top plate 12 can include a variety of stabilizing and strengthening structures, such as gussets, flanges, ribs, and the like, that strengthen the top plate 12 and the flange 13 and provide point(s) of attachment for various components of the fifth wheel 10. For example, a bottom plate 44 is coupled to the top plate 12 and defines a lower surface of the throat 14. The top plate 12 and the flange 13 define a protected space in which operable components of the fifth wheel 10 are positioned.

[0029] An operating arm 20 is pivotally connected to the top plate 12 at a pivot axis 23, and the operating arm 20 is pivotable into and between a locked position (FIG. 1) in which the fifth wheel 10 locks onto the kingpin 16 and an unlocked position (FIG. 2) in which the fifth wheel 10 unlocks from the kingpin 16 (the locked and unlocked positions are described further herein). The operating arm 20 has a first end 21 pivotally coupled to the top plate 12 at the pivot axis 23 via a mechanical fastener such as a pin or bolt. The first end 21 is adjacent to the throat 14. The operating arm 20 is elongated between the first end 21 and an opposite, second end 22. A coil spring 36 biases (e.g., pulls) the operating arm 20 toward the throat 14 in a first direction (see arrow C).

[0030] A pull handle 90 is coupled to the operating arm 20 and is operable to pivot the operating arm 20 from the locked position (FIG. 1) to the unlocked position (FIG. 2). Specifically, the pull

handle **90** is pulled in a second direction (see arrow D) such that the operating arm **20** pivots toward the unlocked position (FIG. 2) and away from the throat **14**. As the operating arm **20** pivots toward the unlocked position (FIG. 2), a wedge **50** and a jaw **60**, which are pivotally coupled to the operating arm **20**, also move away from the throat **14** (see arrow D). Accordingly, the kingpin **16** can be inserted into the throat **14** or removed from throat **14**. When the kingpin **16** is inserted into the throat **14**, the operating arm **20** pivots back to the locked position (FIG. 1), due to the coil spring **36** exerting a pulling force on the operating arm **20**. As the operating arm **20** pivots toward the locked position (FIG. 1), the wedge **50** and the jaw **60** linearly move toward the throat **14** (see arrow C) to thereby lock the fifth wheel **10** onto the kingpin **16** (described further herein). As the wedge **50** and the jaw **60** move toward the throat **14** (see arrow C), the wedge **50** urges the jaw **60** into contact with the kingpin **16** to thereby force the kingpin **16** against a fixed jaw **54** on the top plate **12** and lock the fifth wheel **10** onto the kingpin **16**.

[0031] The wedge **50** is pivotally coupled to the operating arm **20** between the ends **21**, **22** of the operating arm **20** with a wedge pin **51** that is received in a first slot **24** of the operating arm **20**. The wedge pin **51** slides in the first slot **24** as the operating arm **20** pivots such that the wedge **50** linearly moves (see direction arrow E). In certain examples, a knock-out assembly **28** is coupled to the top plate **12** and can be actuated to apply a direct force to the wedge **50** to thereby dislodge the wedge **50** and/or the operating arm **20** in the event either component becomes jammed and prevents the release of the kingpin **16** from the fifth wheel **10**.

[0032] The jaw **60** is also pivotally coupled to the operating arm **20** via a timing lever **70**. The jaw **60** has a jaw pin (not shown) that is received in an elongated slot **76** of the timing lever **70** and a first end **71** pivotally coupled to the operating arm **20** via the wedge pin **51** (see above). As the operating arm **20** pivots, the wedge **50** linearly moves (as described above), the timing lever **70** pivots about wedge pin **51**, the jaw pin slides in the elongated slot **76**, and the jaw **60** linearly moves with the wedge **50**. The timing member **70** has an opposite, second end **72** with a follower pin (not shown) extending therefrom that slides along an outside edge **26** (FIG. 2) of the operating arm **20** as the operating arm **20** pivots. The timing lever **70** is biased toward the throat **14** with an extension spring **78**. The operating arm **20**, the wedge **50**, and the timing lever **70** are all generally plate-like members and are in stacked relation to one another. Reference is made to the above-incorporated U.S. Pat. Nos. 5,641,174 and 5,988,665 for description and operation of a conventional timing lever and associated components.

[0033] A trigger arm **31** is pivotally coupled to the operating arm **20** and is for holding the operating arm **20** in the unlocked position (FIG. 2). The trigger arm **31** extends transverse to the throat **14** and slides on the bottom plate **44** as the operating arm **20** pivots into and between the locked position (FIG. 1) and the unlocked position (FIG. 2). The trigger arm **31** has a trigger **38** that moves into the throat **14** as the operating arm **20** pivots toward the unlocked position (FIG. 2), and the trigger **38** prevents the operating arm **20** from pivoting back to the locked position (FIG. 1) until the kingpin **16** is inserted into the throat **14**. Specifically, when the kingpin **16** is received into the throat **14** the kingpin **16** contacts and moves the trigger **38** out of the throat **14** causing the trigger arm **31** to pivot relative to the operating arm **20** and the trigger **38** to clear the bottom plate **44**. The trigger arm **31** then slides along the bottom plate **44** and the operating arm **20** pivots back toward the locked position (FIG. 1) as the coil spring **36** “pulls” the operating arm toward the throat **14**. The coil spring **36** is shown connected to the trigger arm **31**, however, in other examples the coil spring **36** is directly connected to the operating arm **20**. In the example depicted in FIG. 1, the trigger **38** is a finger member that projects from the trigger arm **31**.

[0034] The operating arm **20** is held or locked in the locked position (FIG. 1) with a secondary lock assembly **80** that is pivotally coupled to the second end **22** of the operating arm **20** via pin **81**. The secondary lock assembly **80** has a pawl member **84** and an opposite, dog member **85** that each radially extend away from a stabilizing pin **83** that is received in an arcuate slot **82** defined in the second end **22** of the operating arm **20**. A coil spring **92**, connected between the secondary lock

assembly **80** and a flange on the top plate **12**, exerts a pulling force in the first direction (see arrow C) to thereby urge the secondary lock assembly **80**, and further urge the operating arm **20**, toward the locked position (FIG. **1**) and in the first direction (see arrow C). In operation, as the operating arm **20** pivots from the unlocked position (FIG. **2**) toward the locked position (FIG. **1**) the secondary lock assembly **80** moves in the first direction (see arrow C) and the pawl member **84** seats behind a stop surface **56** on the top plate **12** to thereby stop or prevent the operating arm **20** from pivoting in the opposite second direction (see arrow D) toward the unlocked position (FIG. **1**). To pivot the operating arm **20** to the unlocked position (FIG. **2**), the pawl member **84** must be pivoted about pin **81** to clear the stop surface **56**. The pull handle **90** is connected to the secondary lock assembly **80** in such a way that as the operator pulls the pull handle **90** in a second direction (see arrow D) the pawl member **84** pivots about pin **81** to clear the stop surface **56** and the operating arm **20** pivots to the unlocked position (FIG. **2**). Accordingly, the kingpin **16** can be received into or moved out of the fifth wheel **10**.

[0035] The present inventors have observed that in certain circumstances the fifth wheel **10** may not fully or properly lock onto the kingpin **16**. For example, the pawl member **84** of the secondary locking assembly **80** may not fully seat behind the stop surface **56** and accordingly, the operating arm **20** does not reach the locked position (see FIG. **1**) and the fifth wheel **10** is in an unlocked operational state. As such, the kingpin **16** could inadvertently move out of the throat **14**. If this occurs on the roadway, the towed trailer may unhitch from the towing vehicle. Accordingly, it is advantageous to provide systems that verify that the fifth wheel **10** is properly locked onto the kingpin **16** and is therefore in the locked operational state. Furthermore, it is advantageous to provide systems that store or log that the operator checked to ensure that the fifth wheel **10** is properly locked onto the kingpin **16** and is in the locked operational state. These systems, as will be described in greater detail below, are capable of sensing and logging proper locking of the fifth wheel **10** onto the kingpin **16** and/or manual interaction between the operator of the towing vehicle and the fifth wheel **10** to thereby create a log that the fifth wheel **10** has been properly locked and/or manually checked or verified by the operator. In addition, the present inventors have also observed that the operable components (e.g., the wedge **50**, the jaw **60**) of the fifth wheel **10** wear over time as the kingpin **16** contacts and rubs on the jaw **60**. This wearing occurs when the kingpin **16** is received into the fifth wheel **10** and during towing. Certain conventional fifth wheels have various “slack” adjustment mechanisms that help account for the wear, however, once wear becomes excessive these mechanisms are no longer able to account for the wear and the operating arm **20** may “over” pivot in the first direction (see arrow C) toward the throat **14**. This additional pivoting may cause vibrations or “jiggling” between the kingpin **16** and the fifth wheel **10** during towing. Accordingly, it is advantageous to provide systems that detect wear of the fifth wheel **10** and that the fifth wheel **10** is in one or more worn operational states. Furthermore, the systems of the present disclosure can alert the operator of the towing vehicle that the fifth wheel **10** is in one of the worn operational states and/or that excessive wear has occurred to one or more components of the fifth wheel **10**.

[0036] As such, the present inventors have developed systems for determining the operational state of the fifth wheel **10** and verifying the operational state of fifth wheel **10**. These systems noted above and further described herein below.

[0037] Referring to FIG. **2**, the system **100** of the present disclosure is shown in relation to the operable components of the fifth wheel **10**, which are described above. In particular, the system **100** includes a magnet **101** on a movable component of the fifth wheel **10** that moves or is movable to lock the fifth wheel **10** to the kingpin **16**. The movable component in this example is the pawl member **84** of the secondary lock assembly **80**, however, a person of ordinary skill in the art will recognized that the magnet **101** can be on any movable component (e.g., the operating arm, trigger arm). One or more sensors **102A-C** on the top plate **12** that are capable of sensing the magnet **101** as the magnet **101** moves past each sensor **102A-C**. Generally, as the operating arm **20** pivots

toward the throat **14** (see arrow C) and the locked position (FIG. 1) the magnet **101** moves past one or more of the sensors **102A-C**. Each sensor **102A-C** that senses the magnet **101** sends a signal or data to a controller **200** which is in communication, via wired or wireless communication links **201**, with the sensors **102A-C**. Based on the data received from the sensors **102**, the controller **200** determines if the operating arm **20** has pivoted into the locked position (FIG. 1) and the fifth wheel **10** is in the locked operational state. The controller **200** can also determine if the operating arm **20** has pivoted past the locked position (FIG. 1) such that the fifth wheel **10** is in a worn operational state, which as is noted above is indicative of excessive wear of operable components of the fifth wheel **10**. Further description of the system **100** is provided hereinbelow.

[0038] Referring now to FIG. 3, a schematic an example system **100** is shown in greater detail. The system **100** includes a printed circuit board (PCB) **205** on which the controller **200** with a memory **202** and a processor **203** (see FIG. 2) and a series of sensors **102A**, **102B**, **102C** are coupled. In this example, the sensors **102A-C** are linearly positioned next to the stop surface **56** (see also FIG. 2) and the magnet **101** is on the pawl member **84** of the secondary lock assembly **80**. The pawl member **84** is shown next to the stop surface **56**, which may occur as the operating arm **20** is pivoting from the unlocked position (FIG. 2) to the locked position (FIG. 1) and moving in the first direction (see arrow C) toward the throat **14** (FIG. 1). When the pawl member **84** is next to the stop surface **56**, none of the sensors **102A-C** sense the magnet **101**. Accordingly, no data is sent to the controller **200** and the controller **200** does not indicate, via an indicator **206** (e.g., operator input/interface device, light emitting diode), that the operating arm **20** is in the locked position (FIG. 2). The fifth wheel **10** in this example is in an unlocked state and the indicator **206** may indicate the unlocked state of the fifth wheel **10**.

[0039] As the operating arm **20** further pivots toward the locked position (FIG. 1) and in the first direction (see arrow C), the pawl member **84** seats behind the stop surface **56** (see also FIG. 4) and the first sensor **102A** senses the magnet **101**. Accordingly, the first sensor **102A** sends data to the controller **200** and the controller **200** controls the indicator **206** to indicate that the operating arm **20** is in the locked position (FIG. 1) and the fifth wheel **10** is in the locked state. FIG. 4 depicts the indicator **206**, which is a multi-color LED, that emits light that is a first color (e.g., green light) when the first sensor **102A** senses the magnet **101**.

[0040] Returning to FIG. 3, if the operating arm **20** continues to pivot in the first direction (see arrow C), the second sensor **102B** senses the magnet **101** and the second sensor **102B** sends data to the controller **200**. As noted above, the operating arm **20** pivots past the locked position (FIG. 1) when excessive wear of the operable components of the fifth wheel **10** is occurring and the fifth wheel **10** is a worn state. As such, the controller **200** indicates, via the indicator **206**, that the operating arm **20** has pivoted past the locked position (FIG. 1) and thereby alerts the operator that the fifth wheel **10** should be inspected and/or repaired. FIG. 5 depicts the indicator **206** emitting light that is a second color (e.g., red light) when the second sensor **102B** senses the magnet **101**.

[0041] Additional sensors, such as the third sensor **102C**, are provided to detect further movement of the operating arm **20** in the first direction (see arrow C) and thereby determine if the operable components of the fifth wheel **10** are additionally worn and the fifth wheel **10** is in other worn states. For example, when the second sensor **102B** senses the magnet **101** and sends data to the controller **200** such that the controller **200** indicates, via the indicator **206**, that the operating arm **20** has moved past the locked position (FIG. 1), the operable components are worn to a first worn state (e.g., 20.0% remaining life), and/or alert the operator that the fifth wheel **10** should be inspected and/or repaired. However, when the second sensor **102B** and the third sensor **102C** sense the magnet **101** (within a predetermined time period) and both send data to the controller **200** the controller **200** indicates, via the indicator **206**, different information to the operator. For example, the controller **200** may indicate that the operable components are worn to a second worn state (e.g. 10.0% remaining life) and/or alert the operator that the fifth wheel **10** should be taken out of service until repaired.

[0042] While FIG. 3 depicts the sensors **102A-C** near the stop surface **56** and the magnet **101** on the pawl member **84**, the sensors **102A-C** and the magnet **101** can be positioned at different locations on the fifth wheel **10** to detect pivoting of the operating arm **20**. For example, the sensors **102A-C** can be placed on the top plate **12** near the first end **21** of the operating arm **20** (see arrow **110** on FIG. 2) in an arc pattern to thereby detect the operating arm **20** as it pivots along an arc path. In this example, the magnet **101** is positioned on the operating arm **20** near the first end **21**.

[0043] The type of sensors **102A-C** may vary and in certain examples are Hall-Effect sensors. In other examples, the sensors **102A-C** are capable of sensing the relative weakness or strength of the magnetic field of the magnet **101**. In addition, while a magnet **101** is described as being sensed by the sensors **102A-C**, the magnet **101** can be replaced with any other suitable element capable of being sensed by the sensors **102A-C**. For example, the sensors **102A-C** may detect the secondary lock assembly **80**, the pawl member **84**, the operating arm **20**, indicia on the operating arm **20**, reflective tape, and/or the like. Furthermore, in certain examples the magnet **101** is coupled to the movable component (e.g., pawl member) of the fifth wheel **10** with a bracket or clip (not shown). In these examples, it is possible to couple the magnet **101** to existing fifth wheel **10** such the existing fifth wheel **10** can be retrofitted to include the system **100**.

[0044] In certain examples, the controller **200** is configured to record and store or log the data received from the sensors **102A-C**. For instance, when the data corresponding to manually checking and/or engagement of the fifth wheel by the operator is received from any one of the sensors **102A-C**, the controller **200** records a timestamp, which can comprise a date and a time, when the data is received. As such, a fleet manager can access this data log to observe operation and wear of the fifth wheel **10**. Furthermore, the data log provides a method for determining if the fifth wheel **10** has been properly cared for and inspected should the towed trailer come unhitched and cause damage and/or other liabilities.

[0045] In certain examples, at least one of the sensors **102** is for sensing a position of the movable component (e.g., pawl member) and the controller **200** is configured to determine the locked state of the fifth wheel **10** based on the position of the movable component. The controller **200** is further configured to monitor the position of the movable component for a predetermined amount of time (e.g., 2.0 minutes, 45.0 seconds) after detecting the locked state. Further, if a threshold change in the position of the movable component is detected by the sensor **101** and thereby determined by the controller **200** within the predetermined amount of time, the controller **200** is further configured to store, on the memory **202** of the controller **200**, a positive indicator of manual verification of the locked state of the fifth wheel **10** by an operator. In certain examples, if the threshold change in the sensed position of the movable component is not detected within the predetermined amount of time, the controller **200** can store a negative indicator of manual verification of the locked state of the fifth wheel **10** by an operator.

[0046] The controller **200** can be on the fifth wheel **10** or remote from the fifth wheel **10**. For example, the controller **200** can be on the control system for the towing vehicle or integral the control system for the towing vehicle such that a separate controller is not needed. The controller **200** and the sensors **102A-C** can be battery powered and/or powered by the power system of the towing vehicle.

[0047] Certain safety rules and/or laws require that the operator of the towing vehicle get out of the towing vehicle to manually and physically check that the fifth wheel **10** is locked onto the kingpin **16** and the fifth wheel **10** is in the locked state. This commonly requires the operator to grasp and shake the pull handle **90** and/or pull the pull handle **90** out a few inches to ensure that the pawl member **84** is seated behind the stop surface **56** (see FIG. 1).

[0048] In other examples, the system **100** can include a handle or secondary sensor **120** (FIG. 2) for sensing movement of the pull handle **90** when the operator physically checks that the fifth wheel **10** is properly locked to the kingpin **16**. The secondary sensor **120** is placed on the pull handle **90** or on the top plate **12**. Once the fifth wheel **10** locks onto the kingpin **16** and the controller **200** logs

one or more data received from the sensors **102A-C**(as described above), the controller **200** is programmed to monitor for data from the secondary sensor **120** within a stored time (e.g. 2.0 minutes, a time period for the operator to get out of the towing vehicle and walk back to the fifth wheel **10**). If the operator engages (e.g. shakes) the pull handle **90** within the stored time, the controller **200** logs the data from the secondary sensor **120** and determines that the fifth wheel **10** was checked by the operator. If no data is received from the secondary sensor **120**, the controller **200** does not record any information.

[0049] Referring to FIG. **6**, an example method for verifying the operational state of the fifth wheel **10** is (FIG. **1**) is depicted. As shown at **602**, the method begins with sensing, with at least one sensor **102A-C**, position of the movable component on the fifth wheel **10** that moves to lock the fifth wheel **10** to the kingpin **16**. The controller **200** is configured to determine if the fifth wheel **10** is in the locked state based on the sensed position of the movable component, depicted at **604**. If the fifth wheel **10** is not in the locked state, the method returns to **602**. However, if the fifth wheel **10** is in the locked state, the controller **200** is configured to monitor the position of the movable component for a predetermined amount of time after determining the locked state of the fifth wheel **10**, depicted at **606**. At **608**, if a threshold change in the sensed position of the movable component is detected within the predetermined amount of time, the controller **200** stores a positive indicator of manual verification of the locked state of the fifth wheel **10** by an operator. The controller **200** may then optionally enter a low-power mode, depicted at **610**, until the controller **200** determines the fifth wheel **10** is not in the locked state. The method then returns to **602**. In certain examples, the threshold change can be a distance (e.g., 1.0 inches of movement from the sensed position).

[0050] Optionally, at **612**, if a threshold change in the sensed position of the movable component is not detected within the predetermined amount of time, the controller **200** stores a negative indicator of manual verification of the locked state of the fifth wheel **10** by an operator. The controller **200** may then optionally enter the low-power mode, depicted at **610**. Still further, the method may optionally include storing a first timestamp when the predetermined amount of time begins and storing a second timestamp when storing a second timestamp when the threshold change occurs, as depicted at **614**.

[0051] In another example, the secondary sensor **120** senses movement of the pull handle **90** out of the fifth wheel **10** as the operator pulls the pull handle **90** and movement into the fifth wheel **10** as the pull handle **90** retracts into the fifth wheel **10**. In this example, a magnet (such as the magnet **101** on the pawl member **84**) is sensed by the secondary sensor **120**. That is, as the pull handle **90** is pulled out of the fifth wheel **10**, the secondary sensor **120** senses the magnet **101** and sends a first signal (e.g., “ON”). Once the secondary sensor **120** does not sense the magnet **101** (due to continued pulling of the pull handle **90** such that the magnet **101** moves past the secondary sensor **120**), the secondary sensor **120** sends a second signal (e.g., “OFF”). When the pull handle **90** is released and the pull handle **90** retracts back into the fifth wheel **10**, the secondary sensor **120** again senses the magnet **101** and sends another first signal. In another similar example, the sensors **102A-C** may sense the magnet **101** as the pull handle **90** moves into and out of the fifth wheel **10** (as described above).

[0052] In still another example, when the pull handle **90** is pulled by the operator at least one of the sensors **102A-C** senses movement of the magnet **101** as the pawl member **84** pivots away from and/or toward the sensors **102A-C**. In this example, at least one of the sensors **102A-C** senses the magnet **101** and sends a first signal (e.g., “ON”) before the pawl member **84** pivots away from the sensors **102A-C**. When the pawl member **84** pivots away from the sensors **102A-C** (due to pulling of the pull handle **90**), at least one of the sensors **102A-C** does not sense the magnet **101** and sends a second signal (e.g., “OFF”). When the pull handle **90** is released, the pawl member **84** pivots, the magnet **101** is moved back toward the sensors **102A-C**, and at least one of the sensors **102A-C** senses the magnet **101** and sends another first signal. In another example, multiple sensors **102A-C** sense the magnet **101** and send different signals as the pawl member **84** pivots. For instance, before

the pawl member **84** pivots away from the sensors **102A-C**, the second sensor **102B** senses the magnet **101** and sends the first signal. At the same time, the first sensor **102A** does not sense the magnet **101** and therefore sends the second signal. As the pawl member **84** pivots away from the sensors **102A-C**, the second sensor **102B** does not sense the magnet and sends the second signal. At the same time, the first sensor **102A** now senses the magnet **101** and therefore sends the first signal. Finally, as the pawl member **84** pivots back toward the sensors **102A-C**(after the pull handle **90** is released) the first sensor **102A** no longer senses magnet **101** and sends the second signal and the second sensor **102B** again senses the magnet **101** and sends the first signal. A person of ordinary skill in the art will recognize that while some of the above examples describe a second signal being sent by the sensor(s), the sensor(s) may not actually send a second signal and instead the controller **200** records absence of the first signal.

[0053] The signals or data received or not received from the sensors **102A-C** and/or the secondary sensor **120** is logged by the controller **200** to thereby provide a detailed log of the operational state of the fifth wheel (e.g., locked state, unlocked state, worn state), presence of the kingpin **16** in the fifth wheel **10**, and/or operator interaction with the fifth wheel **10**. The logged data (e.g., date, time, frequency, locked or unlocked) may be stored locally on the memory **202** of the controller **200** or remotely in the control systems of the towing vehicle, and the logged data can be accessed by the fleet manager. Accordingly, the operator is held accountable for performing all necessary safety checks when operating the towing vehicle and the fifth wheel **10**. Furthermore, the secondary sensor **120** can provide added liability defense for the fleet manager or original equipment manufacturer (OEM). The secondary sensor **120** can be any suitable sensor such as a momentary vibration sensor.

[0054] Referring to FIG. 7, another example system **100** of the present disclosure is depicted. In this example, the sensor **102** senses the magnet **101** as a movable component of the fifth wheel **10**, such as the operating arm **20**, the secondary lock assembly **80**, or the pawl member **84** (FIG. 2), moves to lock the fifth wheel **10** to the kingpin **16**. As discussed in greater detail herein below, the controller **200** receives data from the sensor **102** and processes the data to thereby determine an end position of the movable component after the movable component has moved. For example, the end position may be the position of the pawl member **84** after it seats behind the stop surface **56** (FIG. 2) and stops moving. The controller **200** then compares the end position to a threshold position (described further herein) which may be a position in which the movable component has moved into a position that corresponds to positive locking of the fifth wheel **10** to the kingpin (e.g., the position of the pawl member **84** as depicted in FIG. 1). Accordingly, the controller **200** determines an operational state of the fifth wheel **10**, such as a worn state, a locked state, or an unlocked state, based on the comparison of the end position of the movable component to the threshold position. The controller **200** is coupled to and in communication with an indicator **206** indicates the operational state of the fifth wheel **10** to an operator and/or fleet manager.

[0055] The number and type of operational states of the fifth wheel **10** can vary based on the condition of the fifth wheel **10** and operation thereof. Generally, in the locked state, the fifth wheel **10** is properly locked onto the fifth wheel **10** (FIG. 1). Accordingly, the end position of the movable component, as sensed by the sensor **102**, is in a predetermined locked threshold position stored on the memory **202** of the controller **200** that corresponds to proper movement of components of the fifth wheel **10** and locking of the fifth wheel **10** onto the kingpin **16** (FIG. 1). However, if the fifth wheel **10** does not properly lock onto the fifth wheel, due to incorrect insertion of the kingpin **16** into the throat **14** and/or improper operation of the fifth wheel **10**, the end position of the movable component is not at the predetermined locked threshold position. Therefore, the end position corresponds to an unlocked state of the fifth wheel **10**.

[0056] In addition, as can be appreciated by persons of ordinary skill in the art, stationery and movable components of the fifth wheel **10** may wear over time and thus movable components may move into positions different than a baseline or initial predetermined locked threshold position

when the fifth wheel **10** is locked onto the kingpin **16** (FIG. **1**). Thus, the end position is not in the predetermined locked threshold position, and therefore, the end position corresponds to a worn state of the fifth wheel **10**. One or more predetermined worn threshold positions can be stored on the memory **202** of the controller **200**, and each worn threshold position may correspond to a remaining life expectancy of one or more components of the fifth wheel **10**. For example, a first threshold position corresponds to a first worn state of the fifth wheel in which a component of the fifth wheel **10** has a first remaining life expectancy (e.g., **4000** remaining lock-unlock operations) and a second worn state of the fifth wheel in which a component of the fifth wheel **10** has a second remaining life expectancy (e.g., **500** remaining lock-unlock operations).

[0057] The components of the system **100**, including the sensor **102**, the indicator **206**, the controller **200**, and the other components thereof, are described in greater detail herein below.

[0058] As noted above, the sensor **102** is in operable association with the fifth wheel **10**. Note that in the example depicted in FIG. **2**, the sensors **102A-C** are coupled to the top plate **12**. However, a person of ordinary skill in the art will recognize that the sensor(s), such as the sensor **102** depicted in FIG. **7**, can be connected to any suitable component of the fifth wheel **10**. In one example, the sensor **102** is contained within a water-tight housing (not shown), which is fastened to the top plate **12** via mechanical fasteners or adhesives, so that the sensor **102** is protected from debris and moisture.

[0059] The sensor **102** can be a device capable to sensing magnetic flux generated by the magnet **101** on the movable component of the fifth wheel **10**, such as the pawl member **84** (see FIG. **2**). Furthermore, the specific sensor **102** used in the system **100** may depend on the specific type of magnet **101**. The sensor **102** can be a Hall-Effect sensor. In other examples, the sensor **102** capable of sensing the magnetic field of the magnet **101** in the x, y, and z directions such that the sensor **102** is capable of sensing the three-dimensional movement of the magnet **101** and thereby the movable component of the fifth wheel **10** to which the magnet **101** is coupled. An example of a sensor **102** capable of sensing the three-dimensional movements of the magnet **101** is commercially available from Infineon (model #TLV493D-A1B6). As noted above and depicted in FIG. **2**, more than one sensor may be used in the system **100** (e.g., multiple sensors **102** are used for redundancy and/or error checking other sensors **102**). The sensor **102** generates or outputs position data in the form of analog signals or digital signals, depending on the type of sensor **102** used. In certain examples, the sensitivity of the sensor **102** can be adjusted, either manually or by the controller **200**, to thereby increase the accuracy of the sensor **102** and/or account for variations in the magnetic field that may be affected by the specific location of the magnet **101** on the movable member (e.g., interference of the magnet field caused by certain metallic components of the fifth wheel **10**). In certain examples, the sensor **102** is capable of sensing magnetic flux caused by components of the fifth wheel such that the magnet **101** may be excluded. In other examples, the sensor **102** could be another type of sensor capable of sensing movement of the components of the fifth wheel **10**.

[0060] As noted above, the sensor **102** generates or outputs data to the controller **200** which is configured to process the data. The controller **200** includes the processor **203** and the memory **202**, and the controller **200** can be located anywhere in the system **100**. The controller **200** is in communication with the various components of the system **100** via wired and/or wireless communication links **201**. In certain examples, the system **100** includes more than one controller **200**. The controller **200** includes a timer or counter **210** such that velocity and/or distance traveled can be determined based on the data received from the sensor **102**. The controller **200** is also configured to receive data or inputs from other components in the system **100** such as the operator interface device **220** and/or the indicator **206**. The components of the system **100** (e.g., the controller **200**, the sensor **102**, and the indicator **206**) are powered by a battery **230** and/or a power source (not shown) on the towing vehicle or the towed vehicle.

[0061] As noted above, the controller **200** processes the data to determine an end position of the

movable component based on the magnetic flux caused by the magnet **101**. The controller **200** then compares the end position of the movable component to the locked threshold position such that the operational state of the fifth wheel **10** can be determined. The locked threshold position is predetermined and may correspond to a position in which the pawl member **84** seats behind the stop surface **56** (FIG. 2) when the fifth wheel **10** properly couples to the kingpin **16**.

[0062] In certain examples, the locked threshold position is determined based on controlled, repeatable tests in which the fifth wheel **10** properly couples to the kingpin **16**. Accordingly, the locked threshold position can be identified by examining the end position of one or more movable components of the fifth wheel **10**, and/or the magnet **101** coupled to one of the movable components from each test. The locked threshold position is then inputted into the controller **200** and stored on the memory **202**. In other examples, the controller **200** is configured to “learn” the locked threshold position based on repeated coupling events between the fifth wheel **10** and the kingpin **16**.

[0063] Referring to FIG. 8A, the locked threshold position is on a continuum of potential positions along which the movable component and/or the magnet **101** may move. FIG. 8A is an example linear graphical representation **300** of the potential positions along which the movable component may move during the operation of the fifth wheel **10**. In this example, the sensor **102** generates data corresponding to one coordinate direction (e.g., the x-coordinate) and thereby the controller **200** can determine the position of the movable components along a single coordinate axis (e.g., x-coordinate axis). The linear graphical representation **300** includes a first position extent **301** that may correspond to when the operating arm **20** is in the unlocked position as depicted in FIG. 2. In other examples, the first position extent **301** corresponds to the maximum sensing range of the sensor **102** in a first direction (e.g., in a direction to the right relative to the fifth wheel **10**) along one coordinate axis. The continuum of potential positions extends from the first position extent **301** to a second position extent **302** that may correspond to the maximum sensing range of the sensor **102** in a second direction (e.g., in a direction to the left relative to the fifth wheel **10**) along one coordinate axis. The locked threshold position **303** is predetermined (as noted above) and on the continuum of potential positions between the position extents **301**, **302**.

[0064] In the example depicted in FIG. 8A, the locked threshold position **303** corresponds to the locked state of the fifth wheel **10** in which the fifth wheel **10** is locked onto the kingpin **16**. For instance, the locked threshold position **303** corresponds to when the pawl member **84** is seated behind the stop surface **56** (FIG. 1). In this example, if the controller **200** determines that the end position of the movable component is at the locked threshold position **303**, the controller **200** determines that the operational state of the fifth wheel **10** is the locked state in which the fifth wheel **10** is properly locked to the kingpin **16**. In this example, the controller **200** may also determine the fifth wheel **10** is in the locked state when further movement of the movable component occurs such that the end position is located between the locked threshold position **303** and the second position extent **302** (see position **304**). However, if the controller determines that the end position of the movable component is between the locked threshold position **303** and the first position extent **301**, the controller **200** determines that the operational state of the fifth wheel is the unlocked state in which the fifth wheel is improperly locked or not locked onto the kingpin **16**.

[0065] Referring now to FIG. 8B, the controller **200** can be configured to determine if the operational state of the fifth wheel **10** is in one or more worn states in which one of the components of the fifth wheel is worn. The worn threshold positions are on a continuum of potential positions along which the movable component and/or the magnet **101** may move. FIG. 8B, like FIG. 8A, is an example linear graphical representation **300** of the potential positions along which the movable component may move during the operation of the fifth wheel **10**. Like the example noted above with respect to FIG. 8A, the sensor **102** generates data corresponding to one coordinate direction (e.g., the x-coordinate) and thereby the controller **200** can determine the position of the movable components along a single coordinate axis (e.g., x-coordinate axis). In this example, the controller

200 has one or more predetermined worn threshold positions stored on the memory **202** that are on the continuum of potential positions that extend between the extents **301**, **302**. For instance, a first worn threshold position **311** and a second worn threshold position **312**. The first worn threshold position **311** that corresponds to a first worn state, and the first worn state corresponds a first remaining life expectancy of one or more components of the fifth wheel **10**. (e.g., **4000** remaining lock-unlock operations of the fifth wheel **10**). The second worn threshold position **312** corresponds to a second worn state in which at least one of the components of the fifth wheel **10** is worn, and the second worn state corresponds a second remaining life expectancy of one or more components of the fifth wheel **10**. (e.g., **500** remaining lock-unlock operations of the fifth wheel **10**). By determining the worn state of the fifth wheel **10**, the controller **200** can help the operator and/or fleet manager decide if and/or when the fifth wheel **10** should be scheduled for maintenance, inspected, and/or repaired.

[0066] In this example, if the controller **200** determines that the end position of the movable component is at the first worn threshold position **311**, the controller **200** determines that the operational state of the fifth wheel **10** is the first worn state. However, if the controller **200** determines that the end position of the movable component is at the second worn threshold position **311**, the controller **200** determines that the operational state of the fifth wheel **10** is the second worn state. Note that in certain examples, the controller **200** can be configured to determine the worn state independent from the locked state or the unlocked state (e.g., the controller **200** determines that the fifth wheel **10** is in the work state but does not determine the locked state or the unlocked state). In other examples, the controller **200** can be configured to determine the locked state or unlocked state together with or based on the worn state (e.g., the controller **200** determines that fifth wheel **10** is in the first worn state and therefore, the fifth wheel **10** also in the locked state).

[0067] In certain examples, the controller **200** can be configured to assess position vectors. In this example, the locked threshold position and/or the worn threshold position(s) include two or more positions on the continuum of potential positions. The continuum of potential positions can include positions within a sensing range of the sensor **102**. In addition, the end position determined by the controller **200** includes two or more positions sensed over time as the movable component moves to the locked state. The position vector can include a time series of position measurements that are sensed by the sensor **10** and further processed by the controller **200**. The position vector corresponds to movement of the magnet **101** as the movable component moves to lock the fifth wheel **10** to the kingpin **16** and is based on the data generated by the sensor **102**. In one embodiment, the sensor generates position data relative to two or more coordinates (e.g., the x-coordinate and the y-coordinate) and thereby the controller **200** can determine the position vector of the movable components relative to the two or more coordinate axes (e.g., x-coordinate axis and the y-coordinate axis). The sensed, position vector is then compared to the positions of the locked threshold position and/or the worn threshold position(s).

[0068] Depending on the operational state of the fifth wheel determined by the controller **200**, the controller **200** controls the indicator **206** to thereby indicate the operational state to the operator. The indicator **206** is any suitable indicator, such as a visual indicator (e.g. LED), audio indicator (e.g. speaker), or any other indicator capable of indicating to the operator. Specifically, the indicator **206** may produce an audible alert and/or a visual alert. In certain examples, the indicator **206** is part of the drive system of the towing vehicle. The location of the indicator **206** can vary, such as on the fifth wheel **10** or in the cab on the towing vehicle. The controller **200** may further control the operator input device **220** to thereby display or indicate the operational state to the operator. In operation examples, the indicator **206** is part of the operator interface device **220**, or vice versa.

[0069] Furthermore, the controller **200** may further control the indicator **206** and/or the operator interface device **220** to generate a first alert (e.g., emit yellow light) when the fifth wheel **10** is in the first worn state (as described above) and/or a second alert (e.g., emit red light) when the fifth

wheel **10** is in the second worn state (as described above).

[0070] Referring now to FIG. **9**, an example method for determining the operational state of the fifth wheel **10** (FIG. **1**) is depicted. Note that components of the fifth wheel **10** and/or the system **100** not depicted in FIG. **9** are depicted in FIG. **2**. As shown at **502**, the method begins sensing, with at least one sensor **102**, magnetic flux caused by the magnet **101** on a movable component (e.g., the pawl member **84**) movable to lock the fifth wheel **10** to the kingpin **16** and generating position data that corresponds to the movement of the movable component. Optionally, if the position data generated does not indicate movement of the movable component (e.g., the position data is consistent and does not change because there is no movement of the movable component; the fifth wheel **10** is in the locked state), the controller **200** may enter a low-power mode. Based on the position data received by the controller **200**, the controller **200** determines the end position of the movable component, shown at **506**. At **508**, the controller **200** compares the end position of the movable component to the predetermined locked threshold position and/or the worn threshold position(s) that is stored on the memory **202** (FIG. **6**). Note the threshold position may be part of a look-up table. Based on the comparison of the end position of the movable component to the threshold position, the controller **200** determines the operational state of the fifth wheel, depicted at **310**. The controller **200** then controls the indicator **206** to thereby indicate the operational state of the fifth wheel **10** to the operator, depicted at **312**. Thereafter, the controller **200** may enter the low-power mode (shown at **504**) until the sensor **102** senses additional changes to the magnetic flux caused by the magnet **101** that may be indicative of movement of the movable component and change in operational state of the fifth wheel **10**.

[0071] In certain examples, a method for determining an operational state of a fifth wheel includes sensing, with at least one sensor, magnetic flux caused by a magnet on a movable component movable to lock the fifth wheel to a kingpin of a towed vehicle, determining an end position of the movable component based on the magnetic flux, comparing the end position of the movable component to a threshold position, and determining an operational state of the fifth wheel based on the comparison of the end position of the movable component to the threshold position.

[0072] In certain examples, the method can further include indicating the operational state of the fifth wheel with an indicator. The operational state is at least one of a worn state, a locked state, or an unlocked state. The end position can be on a continuum of potential positions along which the magnet may move as the movable component moves to lock the fifth wheel to the kingpin. The threshold position can be on the continuum of potential positions and corresponds to a worn state of the fifth wheel in which at least one component of the fifth wheel is worn. The method can include generating an alert, with an indicator, when the fifth wheel is in the worn state. In certain examples, the worn state corresponds to remaining life expectancy of the at least one component of the fifth wheel. The worn state can be a first worn state and the method can include comparing the end position to a second threshold position such that the second threshold position corresponds to a second worn state of the fifth wheel in which the at least one component of the fifth wheel is worn and the second worn state corresponds to remaining life expectancy of the at least one component of the fifth wheel that is less than that of the first worn state. In certain examples, the method includes generating, with an indicator, a first alert when the fifth wheel is in the first worn state and a second alert when the fifth wheel is in the second worn state.

[0073] In certain examples, the method includes determining a position vector that corresponds to movement of the magnet as the movable component moves to lock the fifth wheel to the kingpin. The threshold position can include two or more positions on the continuum of potential positions such that determining the operational state includes comparing the magnet position vector to the two or more positions. In certain examples, the end position is determined based on data from at least two sensors. In certain examples, the sensor is a 3D Hall Effect sensor. In certain examples, the threshold position is on the continuum of potential positions and corresponds to a locked state of the fifth wheel in which the fifth wheel is locked onto the kingpin. The method can further

include comprising indicating, with an indicator, if the fifth wheel is in the locked state or an unlocked state. In certain examples, the threshold position includes two or more positions on the continuum of potential positions such that the threshold position is a position vector that corresponds to movement of the magnet as the movable component moves to the locked state of the fifth wheel and the end position includes two or more positions.

[0074] In certain examples, a fifth wheel system includes a fifth wheel configured to couple to a kingpin of a towed trailer and the fifth wheel has a movable component movable to lock the fifth wheel to the kingpin. A magnet is coupled to the movable component, and a sensor is in operative association with the fifth wheel and configured to sense magnetic flux caused by the magnet and generate data corresponding to the magnetic flux as the movable component moves to lock the fifth wheel to the kingpin. A controller is configured to receive the data and process the data to determine an end position of the movable component as the movable component moves, and the controller is further configured to compare the end position of the movable component to a threshold position to thereby determine operational state of the fifth wheel. An indicator indicates the operational state of the fifth wheel.

[0075] In certain examples, the operational state can be at least one of a worn state, a locked state, or an unlocked state. The end position can be on a continuum of potential positions along which the magnet may move as the movable component moves to lock the fifth wheel to the kingpin. In certain examples, the threshold position is on the continuum of potential positions and corresponds to a worn state of the fifth wheel in which at least one component of the fifth wheel is worn. The indicator can generate an alert when the fifth wheel is in the worn state, and the worn state corresponds to remaining life expectancy of the at least one component of the fifth wheel.

[0076] In certain examples, the threshold position is a first threshold position and the worn state is a first worn state such that the controller is further configured to compare the end position of the movable component to a second threshold position that corresponds to a second worn state of the fifth wheel in which the at least one component of the fifth wheel is worn, and the second worn state corresponds to remaining life expectancy of the component of the fifth wheel that is less than the remaining life expectancy of the at least one component of the fifth wheel than the first worn state. In certain examples, the indicator is configured to generate a first alert when the fifth wheel is in the first worn state and a second alert when the fifth wheel is in the second worn state. The controller can be further configured to determine a position vector that comprises the two or more positions as the movable component moves to lock the fifth wheel to the kingpin. In certain examples, the threshold position includes two or more positions on the continuum of potential positions, and controller is configured to compare the magnet position vector to the two or more positions. The end position can be determined based on data from at least two sensors. In certain examples, the sensor is a 3D Hall Effect sensor.

[0077] In certain examples, the threshold position is on the continuum of potential positions and corresponds to a locked state of the fifth wheel in which the fifth wheel is locked onto the kingpin. The indicator can indicate if the fifth wheel is in the locked state or an unlocked state. The controller can be configured to determine a position vector that comprises the two or more positions as the movable component moves to lock the fifth wheel to the kingpin, and the threshold position can include two or more positions on the continuum of potential positions, and wherein controller is configured to compare the position vector to the two or more positions.

[0078] In certain examples, a method of verifying an operational state of a fifth wheel includes sensing a position of a movable component on a fifth wheel movable to lock the fifth wheel to a kingpin of a towed vehicle, determining a locked state of the fifth wheel based on the sensed position, and monitoring the position of the movable component for a predetermined amount of time after detecting the locked state. If a threshold change in the sensed position of the movable component is detected within the predetermined amount of time, the method includes storing a positive indicator of manual verification of the locked state of the fifth wheel by an operator. In

certain examples, if the threshold change in the sensed position of the movable component is not detected within the predetermined amount of time, the method includes storing a negative indicator of manual verification of the locked state of the fifth wheel by an operator. In certain examples, storing a first timestamp when the predetermined amount of time begins and storing a second timestamp when the threshold change occurs.

[0079] In certain examples, a fifth wheel system includes a fifth wheel configured to couple to a kingpin of a towed trailer and the fifth wheel having a movable component movable to lock the fifth wheel to the kingpin. A sensor senses a position of the movable component, and a controller is configured to determine a locked state of the fifth wheel based on the position of the movable component and monitor the position of the movable component for a predetermined amount of time after detecting the locked state and detect a threshold change in the position of the movable component within the predetermined amount of time. The controller then stores a positive indicator of manual verification of the locked state of the fifth wheel by an operator

[0080] In certain examples, a method for determining an operational state of a fifth wheel includes sensing, with at least one sensor, magnetic flux caused by a movable component movable to lock the fifth wheel to a kingpin of a towed vehicle, determining an end position of the movable component based on the magnetic flux, comparing the end position of the movable component to a threshold position, and determining an operational state of the fifth wheel based on the comparison of the end position of the movable component to the threshold position.

[0081] In certain examples, a fifth wheel system includes a fifth wheel configured to couple to a kingpin of a towed trailer and the fifth wheel has a movable component movable to lock the fifth wheel to the kingpin. A sensor is in operative association with the fifth wheel and configured to sense magnetic flux and generate data corresponding to the magnetic flux as the movable component moves to lock the fifth wheel to the kingpin. A controller is configured to receive the data and process the data to determine an end position of the movable component as the movable component moves, and the controller is further configured to compare the end position of the movable component to a threshold position to thereby determine operational state of the fifth wheel.

[0082] Citations to a number of references are made herein. The cited references are incorporated by reference herein in their entireties. In the event that there is an inconsistency between a definition of a term in the specification as compared to a definition of the term in a cited reference, the term should be interpreted based on the definition in the specification.

[0083] In the present description, certain terms have been used for brevity, clarity, and understanding. No unnecessary limitations are to be inferred therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed. The different apparatuses, systems, and method steps described herein may be used alone or in combination with other apparatuses, systems, and methods. It is to be expected that various equivalents, alternatives and modifications are possible within the scope of the appended claims.

[0084] The functional block diagrams, operational sequences, and flow diagrams provided in the Figures are representative of exemplary architectures, environments, and methodologies for performing novel aspects of the disclosure. While, for purposes of simplicity of explanation, the methodologies included herein may be in the form of a functional diagram, operational sequence, or flow diagram, and may be described as a series of acts, it is to be understood and appreciated that the methodologies are not limited by the order of acts, as some acts may, in accordance therewith, occur in a different order and/or concurrently with other acts from that shown and described herein. For example, those skilled in the art will understand and appreciate that a methodology can alternatively be represented as a series of interrelated states or events, such as in a state diagram. Moreover, not all acts illustrated in a methodology may be required for a novel implementation.

[0085] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to make and use the invention. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A method of tracking operational state of a fifth wheel, the method comprising: sensing, with a sensor, movement of a movable component of the fifth wheel that moves as the fifth wheel locks onto a kingpin of a towed vehicle; determining, with a controller having a memory, an operational state of the fifth wheel based on the sensed movement of the movable component; and logging the operational state of the fifth wheel on the memory such that operation and/or wear of the fifth wheel is observable over time.
2. The method according to claim 1, wherein the operational state is at least one of a worn state, a locked state, or an unlocked state.
3. The method according to claim 2, further comprising: determining presence of the kingpin in the fifth wheel; and logging the presence of the kingpin in the fifth wheel to the memory.
4. The method according to claim 2, further comprising generating an alert, with an indicator, when the fifth wheel is in the worn state.
5. The method according to claim 2, further comprising remotely accessing data related to the operational state of the fifth wheel logged on the memory.
6. The method according to claim 1, wherein the sensor is a 3D Hall Effect sensor.
7. The method according to claim 1, further comprising coupling a magnet to the movable component such that the magnet is moved with the movable component, and wherein the sensor is configured to sense the magnetic flux of the magnet such that change in the magnetic flux sensed by the sensor corresponds to the movement of the movable component.
8. The method according to claim 1, wherein the sensor is configured to sense magnetic flux of the movable component such that change in the magnetic flux sensed by the sensor corresponds to the movement of the movable component.
9. The method according to claim 2, further comprising: determining a position of the movable component based on the sensed movement of the movable component; monitoring the position of the movable component for a predetermined amount of time after detecting the locked state; and storing a positive indicator of manual verification of the locked state of the fifth wheel by an operator on the memory when the sensor senses a change in the position of the movable component within the predetermined amount of time.
10. The method according to claim 9, further comprising storing a negative indicator of manual verification of the locked state of the fifth wheel by an operator when the sensor does not sense a change in the position of the movable component within the predetermined amount of time.
11. The method according to claim 9, further comprising: storing to the memory a first timestamp when the predetermined amount of time begins; and storing to the memory a second timestamp when the sensor senses the change in position of the movable component.
12. A fifth wheel configured for coupling to a kingpin of a towed trailer, the fifth wheel comprising: a movable component that moves as the fifth wheel locks onto the kingpin; a sensor in operative association with the fifth wheel and configured to sense movement of the movable component as the fifth wheel locks onto the kingpin and to generate corresponding data; and a controller configured to receive the data, determine an operational state of the fifth wheel based on the data, and log the operational state of the fifth wheel on a memory of the controller such that operation

and/or wear of the fifth wheel is observable over time.

13. The fifth wheel according to claim 12, wherein the operational state is at least one of a worn state, a locked state, or an unlocked state.

14. The fifth wheel according to claim 12, wherein the controller is configured to determine presence of the kingpin in the fifth wheel and log the presence of the kingpin in the fifth wheel to the memory.

15. The fifth wheel according to claim 13, further comprising generating an alert, with an indicator, when the fifth wheel is in the worn state.

16. The fifth wheel according to claim 13, wherein the data stored to the memory can be remotely accessed.

17. The fifth wheel according to claim 12, further comprising a magnet coupled to the movable component such that the magnet is moved with the movable component; and wherein the sensor is configured to sense the magnetic flux of the magnet such that change in the magnetic flux sensed by the sensor corresponds to the movement of the movable component.

18. The fifth wheel according to claim 12, wherein the sensor is configured to sense magnetic flux of the movable component such that change in the magnetic flux sensed by the sensor corresponds to the movement of the movable component.

19. The fifth wheel according to claim 13, wherein the controller is configured to determine a position of the movable component based on the sensed movement of the movable component, monitor the position of the movable component for a predetermined amount of time after detecting the locked state, and store a positive indicator of manual verification of the locked state of the fifth wheel by an operator on the memory when the sensor senses a change in the sensed position of the movable component within the predetermined amount of time.

20. The fifth wheel according to claim 19, wherein the controller is configured to store a negative indicator of manual verification of the locked state of the fifth wheel by an operator when the sensor does not sense a change in the sensed position of the movable component within the predetermined amount of time.
